scenario: A system checks if a user is eligible to vote based on their age.

Logic:

1.Ask the user to enter their age
2.Checks if the age is 18 or older
3. If yes ,print “Eligible to vote.”
4.Otherwise,print “Not eligible to vote.”

---

2. Scenario: A program processes a list of numbers and needs to find the largest value.

Logic:

1. **Read the list of numbers.**
2. **Assume the first number is the largest.**
3. **Iterate through the list, comparing each number with the current largest value.**
4. **If a larger number is found, update the largest value.**
5. **Return the largest number.**

---

3. Scenario: A company provides employees with a 10% bonus if their salary exceeds $50,000.

Logic:

1. **Read the employee's salary.**
2. **If the salary is greater than $50,000, calculate a 10% bonus.**
3. **Otherwise, set the bonus to zero.**
4. **.. Return the calculated bonus amount.**

---

4. Scenario: A program evaluates a number to determine if it is even or odd.

Logic:

1. **Read the input number.**
2. **Check if the number is divisible by 2.**
3. **If yes, print "Even."**
4. **Otherwise print is “odd”**

5.Scenario: A text-processing tool reverses a given word or sentence for formatting purposes

. **Logic:**

1. **input word or sentence.**
2. **Convert the input into a list of characters.**
3. **Reverse the order of characters.**
4. **Join the reversed characters into a string.**

5.Return the reversed word or sentence

---

6. Scenario: A grading system determines whether a student has passed or failed based on their score.

Logic:

1. **Read the student's marks.**
2. **If the marks are 40 or above, print "Pass."**
3. **Otherwise, print "Fail."**

---

7. Scenario: A retail store offers a 20% discount if a customer's total order exceeds $100.

Logic:

1. **Read the total order amount.**
2. **If the amount is more than $100, calculate a 20% discount.**
3. **Subtract the discount from the total amount.**
4. **Return the final amount to be paid.**

---

8. Scenario: A banking system processes withdrawal requests and ensures the user has enough balance.

Logic:

1. **Read the account balance and withdrawal amount.**
2. **If the withdrawal amount is less than or equal to the balance, process the withdrawal.**
3. **Subtract the withdrawal amount from the balance and return the updated balance.**
4. **If the withdrawal amount exceeds the balance, print "Insufficient funds."**

9. Scenario: A calendar system verifies whether a given year is a leap year based on standard leap year rules.

Logic:

1. **Read the input year.**
2. **If the year is divisible by 400, it is a leap year.**
3. **If the year is divisible by 100 but not by 400, it is not a leap year.**
4. **If the year is divisible by 4 but not by 100, it is a leap year.**
5. **Otherwise, it is not a leap year.**

10. Scenario: A program filters out only even numbers from a given list.

Logic:

1. **Read the list of numbers.**
2. **Create an empty list to store even numbers.**
3. **Iterate through the list and check if each number is divisible by 2.**
4. **If divisible, add the number to the new list.**
5. **Return the list of even numbers.**